

CS 4412 M2 Initial Implementation

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I. DATASET DESCRIPTION

My main datasets are the IMDb top 250 movies and Box Office Mojo's top 200 highest grossing movies, with Rotten Tomatoes acting as a bridge between them. Each of my sources represents the main points of my discovery questions; IMDb has the general response, BOM has the profits, and RT has the individual audience and critic responses. An important thing to note about the datasets is that not every movie is shared between them, as in some films were hated but profitable while others became forgotten classics. Instead of fixing this gap in the data, I plan to embrace it. After all, blockbusters and critical darlings are the lifeblood of the industry.

The end result is a catalog of around 415 movies of varying responses and profits. Each movie will have 5 columns. The main three are their box office, their critical reception, and their audience receptions. These categories and the ways they intersect are what I am analyzing for my project. I also have the titles and year of release for the sake of formatting.

Whether I should apply some method to balance the scales, such as calculating the box office in today's money or limiting the data to just the first screening, remains to be seen.

Film Name	Worldwide Box Of	Critical Re:	Audience f	Year of Release
1917	\$446,064,352	88%	88%	2019
2012	\$791,217,826	40%	47%	2009
12 Angry M	\$9,844	100%	97%	1957
12 Years a	\$187,734,091	95%	90%	2013
12th Fail	\$138,288	91%	97%	2023
2001: A Sp	\$68,010,425	90%	88%	1968
3 Idiots	\$60,262,836	100%	93%	2009
A Beautiful	\$316,791,257	74%	93%	2001
A Clockwo	\$27,166,998	86%	93%	1971
A Minecraft Movie	\$961,187,780	47%	84%	2025
A Separati	\$22,926,076	99%	92%	2011
A Silent Vo	\$30,819,442	95%	94%	2016
Aladdin (2019)	\$1,054,304,000	57%	94%	2019
Alice in Wonderla nd (2010)	\$1,025,468,216	51%	55%	2010

Fig. 1. Data Set Fragment

II. DISCOVERY QUESTIONS

The big questions driving my project are:

1. "What kinds of movies emerge based on the critic and audience opinion?"

2. "What mixture of ratings and revenue distinguishes a 'cult classic' from a 'blockbuster'?"

3. "How do good reviews keep a movie alive in the public consensus?"

I wanted to ask these questions because of my extended contact with the media discussion side of the internet, for better or worse. I have seen people claim that a highly profitable film with good critical reviews had bribed reviewers because it had low audience ratings, and I have seen those same people claim that a profitable film with low critical ratings has a cult audience blindly giving the media their money. This inspired me to create the first two questions so I could build a foundation for these opinions to be analyzed. If two movies have the same critic score, but the one that more audiences liked made more money, then that is the true sign of quality.

However, once I took a step back from my personal history, I discovered a very interesting answer to be asked. There is a term known as "Vindicated by History", where a work that failed in the former present becomes heavily praised in the current present. I was reminded of this phrase after I looked through my datasets and found that a sizable number of famous movies don't make that much of a profit. So, I decided to establish another study on the differences between financial impact and cultural impact.

III. EXPLORATORY DATA ANALYSIS

Originally, I wanted to create a punnet square to sort the movies according to their audience and reviewer scores. This would have reflected the Rotten Tomatoes scoring system but not simulate it, as I planned for it to follow my k-means results. However, after going through EDA, I have chosen to follow some more solid graphs.

Once all of the films have been recorded and defined, the real analysis begins. I have a few plots for each column that help establish some findings. The boxplots show that most movies earn under a billion dollars, usually half that, while critics and audiences tend to enjoy what they watch.

Meanwhile, the bar graphs show just how big the gap between profit and popularity really is. The number of low-profit movies compared to everything else, even the ones in the hundreds of millions, combined with the peak reviews being in the good to great range implies the preference of classics over blockbusters. However, my data could have been heavily twisted by outside factors such as the demographic of moviegoers across the years or people's personal biases affecting the total reception.

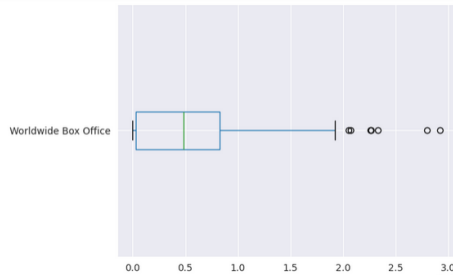


Fig. 2. Box Office Box Plot

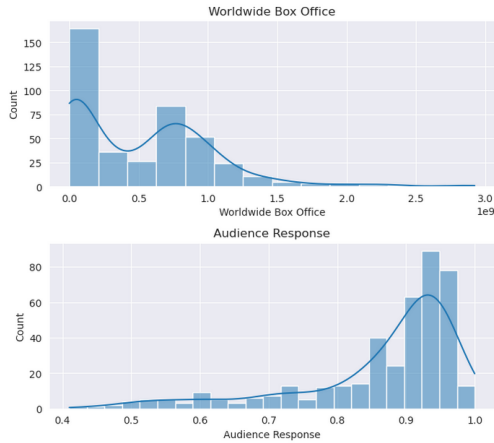


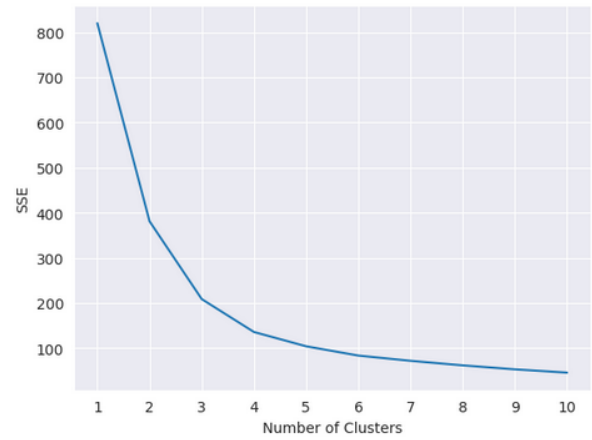
Fig. 3. Bar Graphs for Box Office [Top] and Audience Reception [Bottom]

A. Data Preprocessing

This section has little of note, especially since I ended up deleting or ignoring duplicate values by hand while I was charting my data. As for outliers, the only major one is a 1975 Turkish comedy named *The Chaos Class Failed the Class*, which appears to be a complete anomaly. The minor outliers has a missing box office number or critical reception, but these can be fixed through third party sources.

IV. TECHNIQUES AND FINDINGS

I started by using clustering to mine through my datasets to help smooth down the rough data into my precise rankings. For the sorting of the films, I used k-means to find where the clusters are with a distance matrix of 10. My analysis found three clusters of varying opinions. The first cluster, dubbed 'Genre', is populated by well-received movies with very specialized aesthetics. The more profitable end of the cluster is filled with famous brands like Star Wars and Pixar that bring more people in than their genres. The second, dubbed 'Money Makers', have the films that brought in tons of profit despite some pushback from either audiences or critics. The third and final cluster, dubbed 'Classics', holds movies that everyone remembers due to pop culture osmosis. The age of the data might be why the box office is so low since inflation doesn't apply to them. With these three clusters formed, I can build a more detailed map later on that can analyze a film's impact and legacy.



	Film Name	Worldwide Box Office \
10	Django Unchained	21,900.00
24	12th Fail	138,288.00
37	La Haine	764,368.00
38	Children of Heaven	933,933.00
42	Memories of Murder	1,211,583.00
..	...	...
210	WALL-E	527,404,782.00
211	Star Wars: Episode V - The Empire Strikes Back	550,016,086.00
212	Monsters, Inc.	579,773,055.00
218	The Lost World: Jurassic Park	618,638,999.00
290	Star Wars: Episode IV - A New Hope	775,398,507.00

	Critical Response	Audience Response	Year of Release	cluster
10	0.87	0.92	2012	0
24	0.91	0.97	2023	0
37	0.96	0.95	1995	0
38	0.83	0.95	1997	0
42	0.95	0.92	2003	0
..	...	...	...	...
210	0.95	0.90	2008	0
211	0.93	0.97	1980	0
212	0.96	0.90	2001	0
218	0.57	0.52	1997	0
290	0.94	0.96	1977	0

Fig. 4. K-Means Elbow graph [Top] and Cluster 0 data [Bottom]

V. SOURCES

Github Link

Griffin Vines, "GitHub - PckCalc-Kraftwrk/CS4412-Project," GitHub, 2025. <https://github.com/PckCalc-Kraftwrk/CS4412-Project> (accessed Mar. 05, 2026).

Dataset links

IMDb, "IMDb Top Rated Movies," IMDb, 2019. <https://www.imdb.com/chart/top/>

[2] "Top Lifetime Grosses - Box Office Mojo," Box Office Mojo, 2019. https://www.boxofficemojo.com/chart/ww_top_lifetime_gross/?area=XWW

[3] Rotten Tomatoes, "Rotten Tomatoes: Movies — TV Shows — Movie Trailers — Reviews," Rotten Tomatoes, 2025. <https://www.rottentomatoes.com/>

[4] GeeksforGeeks, "Data Preprocessing in Python," GeeksforGeeks, Oct. 29, 2017. <https://www.geeksforgeeks.org/data-analysis/data-preprocessing-machine-learning-python/>

[5] Zach, "K-Means Clustering in Python: Step-by-Step Example," Statology, Aug. 31, 2022. <https://www.statology.org/k-means-clustering-in-python/>

[6] J. Chen, "Seaborn Swarmplot Axis Interval Formatting," Stack Overflow, Jul. 31, 2020. <https://stackoverflow.com/questions/63196379/seaborn-swarmplot-axis-interval-formatting>